F18CF – Linear Algebra

Tutorial Sheet 10: Stochastic Matrices

Getting started.

1. Determine which of the following stochastic matrices are regular:

$$\begin{pmatrix} 1/2 & 1/3 & 1/2 \\ 1/2 & 1/3 & 0 \\ 0 & 1/3 & 1/2 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1/3 & 1/3 & 0 & 0 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 0 & 0 & 1/3 & 1/3 \end{pmatrix}.$$

2. Compute the general form of the stationary distribution for the following regular stochastic matrices:

$$\begin{pmatrix} 1/2 & 1/4 \\ 1/2 & 3/4 \end{pmatrix}, \quad \begin{pmatrix} 1/2 & 1/4 & 1/2 \\ 1/2 & 1/2 & 0 \\ 0 & 1/4 & 1/2 \end{pmatrix}.$$

Problems.

3. **Migration of population.** A city is trying to model the evolution of its population, following strategic investments. We denote by c_n the number of inhabitants of that city after n years, and by r_n the number of inhabitants in the rest of the country after n years. Under the model used, we have for $n \geq 0$:

$$\begin{pmatrix} c_{n+1} \\ r_{n+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix}}_{=:A} \begin{pmatrix} c_n \\ r_n \end{pmatrix},$$

   (a) Show that the sequences (c_n) and (r_n) converge.

   (b) At equilibrium, what proportion of the population lives in that city?

   (c) What are the eigenvalues λ_1, λ_2 of A? Show that A is diagonalisable.

   (d) Using the fact that the initial condition vector $(c_0, r_0)^T$ can be written in a basis of $\mathbb{R}^2$ made of eigenvectors of A, show that there exists constants $\alpha_1, \alpha_2 \in \mathbb{R}$ such that for every $n \geq 0$:

$$c_n = \alpha_1 \lambda_1^n + \alpha_2 \lambda_2^n.$$

   In this question, we assume that $c_0 = 10^6$ and $r_0 = 10^7$.

   (e) Compute the limit c_∞ of c_n, and find the smallest integer $N \geq 0$ such that for $n \geq N$, we have:

$$|c_n - c_\infty| \leq \frac{1}{100} |c_\infty|.$$

4. **Ranking teams.** (You will need a computer to perform the computations in this exercise) In a sports tournament, five teams $T_1, \ldots, T_5$ are competing against each other. Each team plays against every other team exactly twice. Each game necessarily ends with a winner and a loser (no possibility of a tie). The results of the 20 games are as follows:

   - T_1 lost both games against T_2, and lost both games against T_3.
   - T_5 lost both games against T_1, and lost both games against T_2.
   - T_4 lost both games against T_3.
   - For the remaining pairs of games 'T_i against T_j', T_i won once and T_j won once.

   We wish to adapt the PageRank algorithm to determine the winner of this tournament. We consider a finite Markov chain with five states $T_1, \ldots, T_5$. The probability from going from state T_i to stage T_j is defined as ℓ_{ij}/ℓ_i, where ℓ_i is the total number of games lost by T_i, and ℓ_{ij} is the number of games lost by T_i against T_j.

   (a) Compute the transition matrix of this finite Markov chain, and show that it is regular.

   (b) Adapt the PageRank algorithm (with a damping factor of zero) to determine the winner of this tournament.

   (c) **Food for thought.** Show that the ranking would have been the same for any choice of damping factor $\alpha \in (0, 1)$.